A Comparison of Impermant Loss for Various CFMMs

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***Abstract*—Decentralized exchanges are one of the revolutionary inventions in cryptocurrency trading. Decentralized exchanges present a way to set prices mathematically without maintaining an order book. Order books have been a widely-known method of determining prices for over 400 years in the stock markets and even in decentralized cryptocurrency exchanges. Recently, a new type of decentralized exchange devised a constant function market maker to mathematically determine prices. Among them, constant product market makers are the most widely used. In addition, several constant function market makers have been proposed. However, there was no discussion about the desirable requirements of a constant function market maker in terms of impermanent loss. In this paper, we discuss the desirable requirements of a constant function market maker and analyze the impermanent loss of various market makers. Note that two market makers, LMSR and CCMM, show impermanent gain in the mathematical derivation, but their price functions violate the principle of supply and demand. Therefore, AMMs with impermanent gain while following the principle of supply and demand need to be studied further. A comparison study of impermanent loss for some CFMMs shows interesting results, too.**

Keywords—blockchain, cryptocurrency, decentralized exchange, impermanent loss, permanent loss, market maker, smart contract

I. Introduction

Order book-based trading systems have a history of more than 400 years in the stock market. Centralized cryptocurrency exchanges have also adopted order book-based trading systems. Early decentralized exchanges such as EtherDelta over Ethereum were also order book-based because there were no significant alternatives. Maintaining an order book through a smart contract on the Ethereum blockchain is costly and tends to be slow due to a complicated matching mechanism of order books, and the limited capacity of the blockchain [1]. Later, newly emerging decentralized exchanges started to use automated market makers (AMMs), which do not require order books for trading.

AMMs have shown that exchanges without order books can be utilized for trading, which is a remarkable innovation because there is no price discovery process, where a proper price must be discovered between sellers and buyers in order book-based trading systems. AMM decides the asset price using a mathematical function regardless of what sellers and buyers want. Traders can decide only when and how much to trade. There has never been a coercive method of price determination like this in the past, but it has been used well in the market. AMM's smart contract determines the price immediately with no exceptions according to pre-defined rules and it leads to a sense of fairness in trading. In the decentralized exchanges with AMMs, there are traders and liquidity providers who facilitate smooth trading. The counterparty of traders in AMM is not a human but a smart contract. Even if AMM traders have losses, the smart contract makes no profit. However, whether AMM traders make a profit or a loss, liquidity providers always suffer a loss. Such an unavoidable loss to liquidity providers is called *impermanent loss*, which is one of the most significant research topics in decentralized finance [2][3][4][5][6][7][8]. Among those, most studies were devoted to impermanent loss in CPMM (constant product market maker)-type AMM. It is important to note that almost half of liquidity providers are losing money due to impermanent loss [9].

Bancor [10] was the first decentralized exchange that introduced a liquidity pool smart contract on the Ethereum network in 2017 [11]. The liquidity pool smart contract allows traders to buy and sell assets immediately at any time since each liquidity pool consists of a pair of base and pricing assets deposited. For example, Ethereum (ETH) is the base asset to be traded, and USDT or USDC is the pricing asset (a.k.a. stablecoin) in the liquidity pool. The price of the base asset is set mathe-

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matically by the smart contract, irrespective of the intention of the buyer or seller. Thus, asset trading can be completed in an instant due to the abundant liquidity and mathematical pricing.

A few years before Bancor, Hanson [12][13] introduced LMSR (logarithmic market scoring rule) for the prediction market. The LMSR can be used as an automated market maker [14]. Othman *et al.* [15] proposed LS-LMSR (liquidity-sensitive logarithmic market scoring rule), which is an advanced version of LMSR. However, the LMSR-type AMMs require intensive computations to determine prices through smart contracts and have several other deficiencies.

Uniswap [16] adopted CPMM (constant product market maker) as AMM and showed that it is sufficiently popular, effective and sustainable, leading the decentralized exchanges. Later, Balancer [17] introduced the CMMM (constant mean market maker) that generalized the CPMM with less impermanent loss. In Wang's study [14], CSMM (constant sum market maker) and CCMM (constant circle market maker) are introduced. Recently, Angeris and Chitra [17] introduced a CFMM (constant function market maker) model as a generic model covering all of them.

Impermanent loss can be considered as a penalty imposed on liquidity providers. It occurs in a situation where liquidity providers benefit more from not depositing assets than depositing them. Most liquidity providers can still provide liquidity despite impermanent loss due to interest and governance token rewards for them. In addition, most AMMs have been tried and consistently failed in avoiding impermanent loss. Recently, Kim *et al.* [6] computed the impermanent loss of four automated market makers under the assumption that the price of an asset in a liquidity pool is the ratio of the number of two assets and showed that impermanent gain is achievable in certain cases. However, this assumption is effective only for CPMM market makers. As Kim *et al.* [21] showed, the only case where the price is expressed as a ratio of the numbers of two assets is the CPMM of Uniswap v1 and it makes the impermanent gain revealed in Kim *et al.*'s paper [6] unlikely, to exist.

In this paper, we derive an impermanent gain in LMSR and CCMM. This outcome has never been reported so far to our best knowledge. However, these two CFMMs also show unrealistic characteristics that the price function violates the principle of supply and demand. Based on the principle of supply and demand, when the quantity of an asset increases, the price should fall, and when the quantity decreases, the price should rise. We also have investigated a variant of CCMM and show it follows the principle of supply and demand.

The mathematical price functions used in this paper were derived by Kim *et al.* [21]. Based on this price function, we investigate the value gain of various CFMMs. It is widely known that CPMM and CMMM are CFMMs with impermanent loss. On the other hand, we show that LMSR and CCMM are CFMMs with impermanent gain. However, these CFMMs do not follow the principle of supply and demand while their cost functions are opposite of CPMM and CCMM in terms of convexity [21]. It implies that AMMs should be investigated further. To make the CFMMs follow the principle, a slight modification is conducted, and their outcomes are promising. A comparison study of impermanent loss for CFMMs shows that impermanent loss can be controlled.

II. Mathematical Price and Value Gain

In this paper, all AMMs consider only a constant cost function containing two variables x and y. The constant cost function such as $C(\mathbf{q}) = k$ is called constant function market maker, where k is a constant. A state $\mathbf{q}$ has two variables x and y, representing the number of two outstanding tokens.

Hanson's LMSR automated market maker [12][13] was originally proposed for the prediction markets. His price concept meets the requirements for prediction markets. Since his prices represent the probability of winning in the prediction, the sum of all prices always has to be 1. Othman *et al.* defined this property as translation invariance [15]. The price p_x of x and the price p_y of y are obtained by differentiating the cost function as follows.

$$p_x = \frac{dC(\mathbf{q})}{dx} \quad \text{and} \quad p_y = \frac{dC(\mathbf{q})}{dy}.$$

When the sum of the two prices is 1 (i.e. $p_x + p_y = 1$), the price is defined as translation invariant. Therefore, since the price of the stablecoin is already set at 1, it is unreasonable to use this price on the exchange.

Automatic market makers of decentralized exchanges use a different pricing system. As the price of the stablecoin is set at 1, so P_y is 1. P_x is determined as the ratio of the difference between x_{t-1} and x_t and the difference between y_{t-1} and y_t. In other words,

$$P_x(\mathbf{q_t} - \mathbf{q_{t-1}}) = -\frac{y_t - y_{t-1}}{x_t - x_{t-1}}.$$

If the difference between x_{t-1} and x_t is infinitesimal, we obtain

$$P_x(\mathbf{q_t}) = \lim_{x_t \to x_{t-1}} -\frac{y_t - y_{t-1}}{x_t - x_{t-1}}$$

$$= -\frac{dy_t}{dx_t} = \frac{p_x}{p_y}.$$

The value of the assets in the automated market makers is given as a product of asset price and number as follows:

$$V_x = P_x \cdot x \quad \text{and} \quad V_y = y.$$

Impermanent loss is a term created on the premise that there is no time when the value of the liquidity provider's assets is a profit. However, the terminology impermanent loss, makes it difficult for us to hope that permanent or impermanent gain will ever exist. Thus, this paper introduces a neutral definition of liquidity provider's *value gain*, G, as follows:

$$G_t = P_x \cdot (x_t - x_0) - (y_t - y_0).$$

where x_0 and y_0 refer to the quantity of the base asset and the pricing asset at the time the liquidity provider first entrusted the asset pair. If $G_t \leq 0$, then the liquidity provider suffers from impermanent loss.

III. LMSR AND LS-LMSR

Hanson proposed the LMSR market maker [12][13] which has the cost function as follows:

$$C(\mathbf{q}) = b \cdot \ln\left(e^{\frac{x}{b}} + e^{\frac{y}{b}}\right) = k$$

for constant b and state $\mathbf{q} = (x, y)$, where x and y represent the numbers of two outstanding tokens, respectively. Its price function is given as follows:

$$P_x(\mathbf{q}) = \frac{p_x}{p_y} = \frac{e^{\frac{x}{b}}}{e^{\frac{y}{b}}} = \frac{e^{\frac{x}{b}}}{e^{\frac{k}{b}} - e^{\frac{x}{b}}}.$$

The value of each asset in the LMSR automated market maker is given as follows:

$$V_x = \frac{e^{\frac{x}{b}}}{e^{\frac{k}{b}} - e^{\frac{x}{b}}} \cdot x,$$

and

$$V_y = y = b \ln\left(e^{\frac{k}{b}} - e^{\frac{x}{b}}\right).$$

Notably, impermanent gain is possible in the case of LMSR simulation. For $x_0 = 100, y_0 = 1000$, and $b = 10{,}000$, Figure 1 shows that $G \geq 0$ for all positive x_t. However, it is shown in Figure 1 that the price function does not follow the principle of supply and demand. The principle states that if supply increases while demand remains constant, price decreases, and if supply decreases while demand remains constant, price increases. Therefore, further study and discussion need to be conducted to determine whether LMSR is suitable for AMM trading systems. If everyone agrees that LMSR is a suitable trading method for AMM, this study discovers that LMSR guarantees permanent gain.

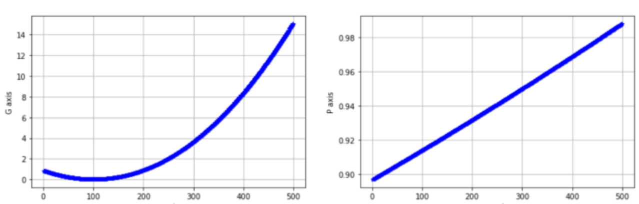

Fig. 1. Value gain and price relationship vs. asset supply of LMSR CFMM for $x_0 = 100, y_0 = 1000$, and $b = 10{,}000$.

We conjecture that LS-LMSR may have a similar pattern as LMSR. However, LS-LMSR is not covered in this paper. Note that the LMSR satisfies the principle of supply and demand when the parameter b is negative. In this case, the LMSR produces the impermanent loss curve. Figure 1 shows the impermanent gain since the parameter b is positive.

IV. CPMM

CPMM cost function, which is used in Uniswap [1], is a product of x and y as follows:

$$C(\mathbf{q}) = x \cdot y = k.$$

Then,

$$p_x = \frac{dC(\mathbf{q})}{dx} = y, \qquad p_y = \frac{dC(\mathbf{q})}{dy} = x.$$

Thus,

$$P_x(\mathbf{q}) = \frac{p_x}{p_y} = \frac{y}{x} = \frac{k}{x^2}.$$

Impermanent loss of CPMM is a well-known fact. Figure 2 shows that impermanent loss is obvious, and the price follows the principle of supply and demand.

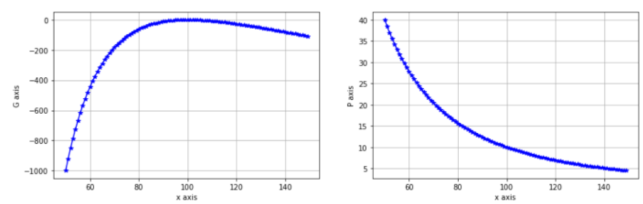

Fig. 2. Value gain and price relationship vs. asset supply of CPMM v1 CFMM for $x_0 = 100$ and $y_0 = 1000$.

Uniswap V3 uses constrained liquidity, such as

$$\left(x + \frac{k}{\sqrt{P_b}}\right)\left(y + k\sqrt{P_a}\right) = k^2.$$

This CPMM is effective for

$$0 \leq x \leq k\frac{\sqrt{P_b} - \sqrt{P_a}}{\sqrt{P_a} \cdot \sqrt{P_b}}$$

and its price range is limited to

$$P_a \leq P_x(\mathbf{q}) \leq P_b.$$

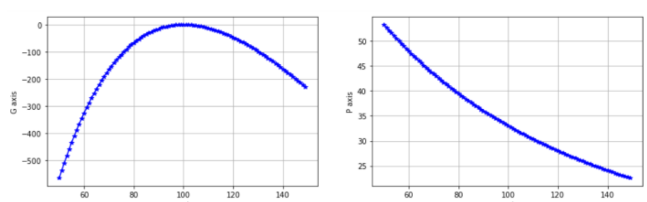

Fig. 3. Value gain and price relationship vs. asset supply of CPMM v3 CFMM for $x_0 = 100$, $y_0 = 1000$, $P_a = 25$, $P_b = 100$ and $k = 1348$.

The price of the Uniswap V3 market maker is represented as follows:

$$P_x(\mathbf{q}) = \frac{p_x}{p_y} = \frac{y + k\sqrt{P_a}}{x + \frac{k}{\sqrt{P_b}}} = \frac{k^2}{\left(x + \frac{k}{\sqrt{P_b}}\right)^2}$$

Figure 3 shows a similar result as Figure 2 since both curves are from CPMM. Figure 3 is the result of the Uniswap V3 (i.e., constrained liquidity). It has been observed that Uniswap v3 has more severe impermanent loss than Uniswap v1.

V. CSMM

CSMM cost function cannot be used as an automated market maker because its price function is constant no matter what the state is. Its cost function is given as follows:

$$C(\mathbf{q}) = x + y = k.$$

Its price function is given as follows:

$$p_x = 1 \text{ and } p_x = 1.$$

Thus,

$$P_x(\mathbf{q}) = \frac{p_x}{p_y} = 1.$$

Therefore,

$$G_t = P_x \cdot (x_t - x_0) - (y_t - y_0) = 0.$$

Therefore, it is well known that CSMM gives neither permanent loss nor permanent gain, and the value gain is always zero.

VI. CMMM

CMMM is currently being used in a single decentralized exchange, Balancer [17]. CMMM's cost function has one desirable feature. CMMM can be seen as better than CPMM in terms of impermanent loss. CMMM has a milder degree of impermanent loss than CPMM. CMMM cost function is given as follows:

$$C(\mathbf{q}) = x^\alpha \cdot y = k.$$

Thus, CPMM is a special case of CMMM with α = 1.

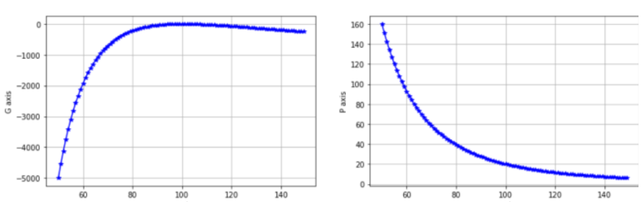

Fig. 4. Value gain and price relationship vs. asset supply of CMMM CFMM for $x_0 = 100$ and $y_0 = 1000$.

For CMMM, the price function is given as follows:

$$p_x = \alpha x^{\alpha-1} \cdot y, \qquad p_y = x^\alpha,$$

and

$$P_x = \frac{\alpha x^{\alpha-1} \cdot y}{x^\alpha} = \alpha \frac{k}{x^{\alpha+1}}.$$

Based on the price function, we can compute the value gain and confirm that CMMM is better than CPMM in terms of impermanent loss, as shown in Figure 4.

VII. CCMM

The CMMM cost function is given as follows:

$$C(\mathbf{q}) = x^2 + y^2 = k^2.$$

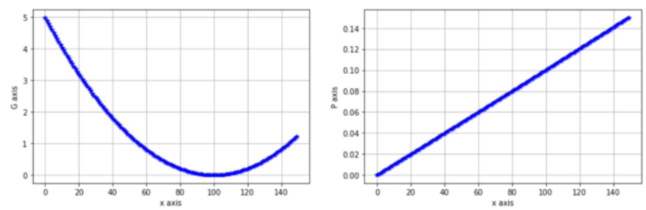

Fig. 5. Value gain and price relationship vs. asset supply of CCMM CFMM for $x_0 = 100$ and $y_0 = 1000$.

This cost function is effective only in the non-negative range as follows:

$$0 \leq x, y \leq k.$$

In this case, the CCMM automated market maker is concave and the price function is derived as follows:

$$p_x = 2x, \qquad p_y = 2y,$$

and

$$P_x = \frac{x}{y} = \frac{x}{\sqrt{k^2 - x^2}}.$$

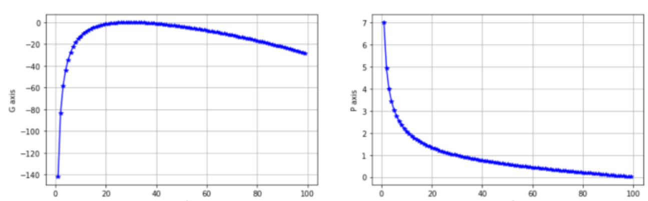

Fig. 6. Value gain and price relationship vs. asset supply of a translated CCMM CFMM for $x_0 = y_0 = 100 \cdot \frac{\sqrt{2}-1}{\sqrt{2}}$.

Based on the price function, we can compute the value gain and confirm that CCMM shows the impermanent gain, as shown in Figure 5. Figure 5 shows that CCMM price also violates the principle of supply and demand. Let's consider a variant form of CCMM. The translated CMMM cost function with its center at (*k, k*) is given as follows: as follows:

$$C(\mathbf{q}) = (x - k)^2 + (y - k)^2 = k^2.$$

In this case, the price function is derived as follows:

$$p_x = 2(x-k), \qquad p_y = 2(y-k),$$

where

$$y = k - \sqrt{k^2 - (x-k)^2}$$

and

$$P_x = \frac{x-k}{y-k} = -\frac{x-k}{\sqrt{k^2-(x-k)^2}}.$$

In this case, Figure 6 shows that the variant CCMM shows a similar pattern to CPMM and CMMM. With the parameters $k = 100$, and

$$x_0 = 100 \cdot \frac{\sqrt{2}-1}{\sqrt{2}}, \qquad y_0 = 100 \cdot \frac{\sqrt{2}-1}{\sqrt{2}}$$

Figure 6 shows that the price function follows the principle of supply and demand, and in this case, obviously impermanent loss is observed.

Note that y should be between 0 and k. Thus, we have to discard the following y since

$$y = k + \sqrt{k^2 - (x-k)^2} \geq 0.$$

VIII. Comparison of Cost Functions

Various AMM cost functions and their price functions are derived in the previous sections. In this section, impermanent loss curves are compared. As mentioned in the previous sections, the LMSR with the positive parameter b and the CCMM with its center at the origin $(0,0)$ are not suitable as a candidate for AMM since they violate the principle of supply and demand. Therefore, the LMSR with negative b, the CPMM, the CPMM with constrained liquidity, the CMMM, and the translated CCMM with its center at (k,k) are compared.

For each CFMM, it is important to choose an appropriate values of k, which may be used to control impermanent loss. Note the value of k should be the positive value for all CFMMs in this paper. Since the value of k characterizes the liquidity pool, it is important to choose the value of k carefully. Given the initial value of liquidity pool (such as $x_0 = 100$ and $y_0 = 1000$ for fair comparison). Then, it is straightforward to obtain the values of k from the cost functions directly, except in the case of the translated CCMM with its center at (k,k). For the translated CCMM with its center at (k,k), the condition to choose a pair of valid initial values can be derived from the following cost function

$$C(\mathbf{q}) = (x_0 - k)^2 + (y_0 - k)^2 = k^2$$

as follows:

$$k = (x_0 + y_0) \pm \sqrt{2x_0 y_0} > 0.$$

It is obvious that

$$(x_0 + y_0) > \sqrt{2x_0 y_0}$$

for all positive x_0 and y_0. Thus, the initial values $x_0 = 100$ and $y_0 = 1000$ are valid and the values of k are

$$k = 1100 \pm \sqrt{200000}.$$

Since there are two valid values of k, we have to conduct a comparison twice with $k = 1547.21$ and $k = 652.78$.

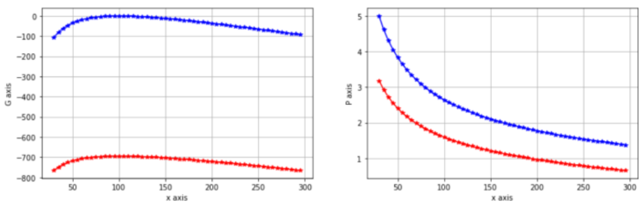

Fig. 7. Value gain and price relationship vs. asset supply of a translated CCMM CFMM for $x_0 = 100$ and $y_0 = 1000$ with $k = 1547.21$ (in blue) and $k = 652.78$ (in red).

Figure 7 shows that the translated CCMM shows either an impermanent loss (in the blue curve) or permanent loss (in the red curve) with different values of k with the same initial value. The permanent loss case is identified first time in this paper. The permanent loss is more severe than the impermanent loss in Figure 7. Therefore, liquidity providers will not prefer the permanent loss case.

Now, CPMM and the translated CCMM are compared with the same initial values $x_0 = 100$ and $y_0 = 1000$. Then, the value of k for CPMM is 10,000 while that for the translated CCMM is 1547.21.

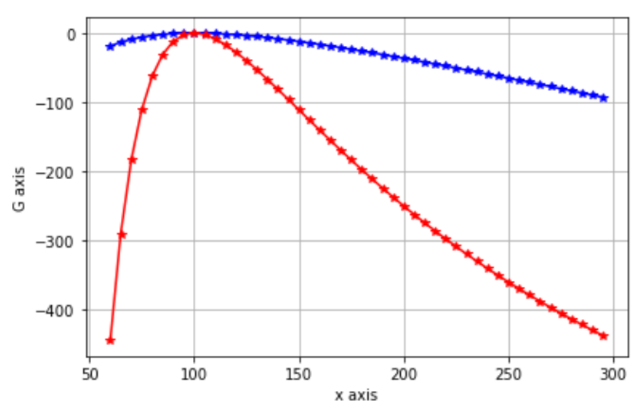

Fig. 8. Value gain vs. asset supply of a translated CCMM with $k = 1547.21$ (in blue) and the CPMM for $x_0 = 100$ and $y_0 = 1000$ (in red).

Figure 8 shows that the impermanent loss of the CPMM is more severe than the translated CCMM. Thus, the translated CCMM is more attractive than the CPMM in terms of impermanent loss.

Now, the CPMM is compared with the CMMM with $\alpha = 2$ and the CMMM with $\alpha = 1/2$. It is a well-known fact that

CMMM shows an improved performance in terms of impermanent loss compared to the CPMM with $\alpha > 1$ [22]. The green line in Figure 9 is the impermanent loss curve with $\alpha = 2$. On the other hand, the blue curve represents the impermanent loss of the CMMM with $\alpha = 1/2$. It is clear that the CMMM with $0 < \alpha < 1$ produces a worse impermanent loss curve than the CPMM.

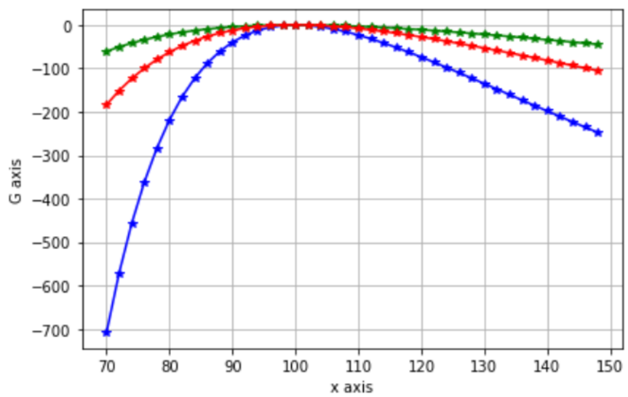

Fig. 9. Value gain vs. asset supply of the CPMM (in red), the CMMM with $\alpha = 2$ (in green) and the CMMM with $\alpha = 1/2$ (in blue) for $x_0 = 100$ and $y_0 = 1000$.

The LMSR with a negative b is compared to the CPMM. For the comparison, we set the value of b to -1000. Figure 10 shows that the LMSR in blue color is better than the CPMM in red color.

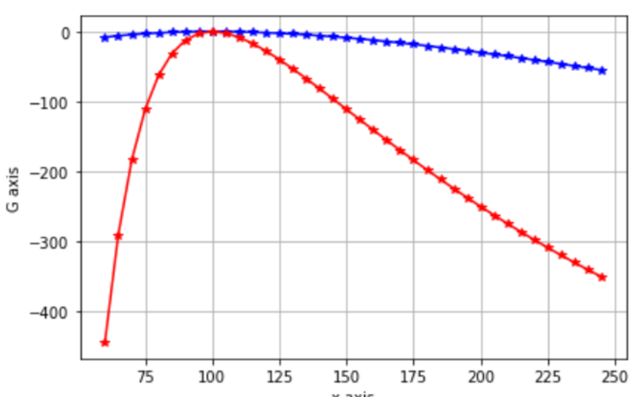

Fig. 10. Value gain vs. asset supply of the CPMM (in red) and the LMSR with $b = -1000$ (in blue) for $x_0 = 100$ and $y_0 = 1000$.

From Figures 10 and 11, we can see that the LMSR with a proper value of b controls the performance of impermanent loss. We can choose the best CFMM and select the most proper value of the parameter to achieve better impermanent loss performance.

IX. Conclusion

In this paper, we examined well-known automated market makers, including LSMR, LS-LMSR, CPMM, CSMM, CMMM, and CCMM. Each market maker has its pros and cons. In order to understand the pros and cons, we introduced a concept of value gain and observed the value gains of AMM methods mentioned above by using the price functions [21].

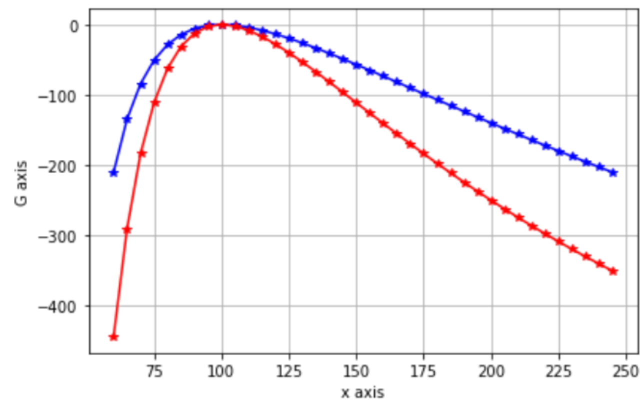

Fig. 11. Value gain vs. asset supply of the CPMM (in red) and the LMSR with $b = -490$ (in blue) for $x_0 = 100$ and $y_0 = 1000$.

The cost functions of LMSR with a positive parameter value of b, LS-LMSR, and the CCMM are concave, and their price functions violate the principle of supply and demand. Their price functions are monotonically increasing with respect to x. In order to make the price function satisfy the principle of demand and supply, as an alternative, we suggest the reciprocal of the monotonically increasing price as a desirable price. On the other hand, the cost function of the LMSR with negative b, the CPMM and the constrained liquidity CPMM, the translated CCMM, and CMMM are convex, and their price function is monotonically decreasing.

CCMM can be either convex or concave. Based on this fact, a variant of CCMM is used to make a CFMM following the principle of supply and demand. The translated CCMM does. However, when the CFMM follows the principle of supply and demand, impermanent loss is unavoidable. Since the CPMM and CMMM CFMMs are widely used in the markets, neither LMSR nor CCMM received attention. To avoid impermanent loss and achieve impermanent gain, new AMMs similar to the LMSR with negative parameters and the translated CCMM need to be proposed and studied further.

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